

Zhihan Chen

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Master of Engineering in Financial Engineering, GPA: 3.62/4.0

Expected Dec. 2025

University of California, Los Angeles, Los Angeles, CA
Bachelor of Science in Applied Mathematics, GPA: 3.87/4.0

Sept. 2022 to March 2024

University of California, Santa Barbara, Santa Barbara, CA
Financial Mathematics, GPA: 3.94/4.0

Sept. 2020 to August 2022

Selected Coursework: Financial Engineering with Stochastic Calculus, Optimization, Machine Learning, Differential Equations, Linear Algebra, Numerical Methods, Mathematical Analysis, Probability and Statistics, Fixed Income Analysis, Stochastic Process, Time Series Analysis, and Portfolio management

SKILLS

Technical: Python, C++

EXPERIENCE

Quantitative Investment Researcher, *Tonghuashun Technology*, Financial Research, Hangzhou, China **May to Aug. 2024**

- Conducted statistical and economic research for securities research report to replicate and optimize quantitative factors which reflect market signal; leading to an optimized back-testing return of 15% with Information Ratio of 3.02, turnover ratio of 3.57 and transaction cost 1%.
- Conducted data analysis on China Stock Market, and worked closely with the project manager to monitor daily stock change signals and patterns, including mean, std, kurtosis, skewness, and annualized return.
- Implemented ETF Fund Rotation Systematic strategies based on buy-sell ratio, achieving a maximum 4 year cumulative return of 80%, sharp ratio 1.2, and maximum drawdown 17%.

Fixed Income Trader, *Caitong Fund*, Trading, Shanghai, China

June to Sept. 2023

- Assisted the trader to verify the transaction data of the pledged style repo, including the direction of the party, the name of counter-party, the transaction amount, and the repo term. Ensuring there is no error made during the bond transaction.
- Used Excel to organize data and applied function to analyze (sort, classify, arithmetic calculation) the transaction data.

PROJECTS

"Emotion Classification of Natural Language" Machine Learning Project, *Cornell University* **Aug. to Dec. 2024**

- Trained SVM and Naive Bayes models to recognize human emotions expressed in piece of text with best SVM validation accuracy 0.7258 and best Naive Bayes accuracy 0.5842.
- Applied BERT model as part of feature engineering and data processing process to optimize the SVM model and successfully raised the accuracy to 0.77.

"Gone With the Wind" Mathematical Modeling, *UCLA*

Sept. to Dec. 2023

- Proposed a new Ordinary Differential Equation system to quantify the love dynamic between hero and heroine in *Gone with the Wind*. Using programming language to draw the phase portrait and trajectory of the new model.
- Optimized the model by splitting single terms in the original model into multiple linear combinations and reassigning the values of the terms within these linear combinations.

LEADERSHIP EXPERIENCE

Event Planner, *UCLA Undergraduate Mathematics Student Association*, Los Angeles, CA

Sept. 2022 to June 2023

- Assisted 10+ students with Math problems on differential equations and linear algebra.
- Organized 5 meetings with UCLA Alumni and asked them to share research and industry experience with over 30 attendees each meeting.

ACTIVITIES/INTERESTS

Activities/Interests: Guitar, Basketball, Fitness, Badminton, Hiking